

1. **Discuss the merits and demerits of river water source (Melamchi khola) and ground water source (in Kathmandu valley) for water supply scheme in Kathmandu.**

Answer:

For Kathmandu Valley, **Melamchi Khola** and **groundwater** are both important sources. The Melamchi Water Supply Project is designed to divert about **170 MLD** of water to the Valley through a 26 km tunnel, making it an important surface-water source for long-term supply.

Merits of river water sources:

- i. Melamchi Khola provides a relatively large quantity of surface water and can meet a significant portion of Kathmandu's growing demand.
- ii. The project is designed to convey about **170 MLD**, with provision for future augmentation from the Yangri and Larke rivers.
- iii. Use of Melamchi water reduces pressure on the over-exploited groundwater resources of Kathmandu Valley.
- iv. Surface water from a properly managed watershed can provide a more sustainable source for a large urban population than continuously pumping groundwater.
- v. Additional water can potentially be brought from the nearby Yangri and Larke river systems.

Demerits:

- i. Intake, treatment plant, tunnel, transmission pipelines and distribution infrastructure require very large capital investment.
- ii. Water has to be conveyed through a tunnel of about **26 km**, increasing construction, operation and maintenance requirements.
- iii. River discharge and water quality vary considerably between wet and dry seasons.
- iv. Diversion of water can affect downstream users and aquatic ecosystems if environmental flows and competing water uses are not properly managed.
- v. Damage to the intake, tunnel or transmission system can disrupt a large centralized supply system.

Merits of groundwater sources:

- i. Groundwater is available within the Valley, so long transmission systems are generally unnecessary.
- ii. Wells, tube wells and pumping systems can be developed incrementally according to demand.
- iii. Soil and aquifer materials provide some natural filtration, so groundwater is generally less turbid than river water.
- iv. Properly managed aquifers provide storage and can supply water during dry periods.
- v. Groundwater can effectively supplement surface-water supplies, particularly during dry seasons.

Demerits

- i. Groundwater extraction in Kathmandu Valley has exceeded recharge in many areas, causing declining groundwater levels.
- ii. Excessive pumping has contributed to substantial groundwater-level decline and drying of traditional water sources such as stone spouts.
- iii. Groundwater is a finite resource; pumping cannot sustainably exceed natural and artificial recharge.
- iv. Deep wells require electricity for pumping, increasing operating costs.
- v. Excessive groundwater withdrawal can contribute to environmental degradation and potentially land subsidence.

2. **Describe the different types of living organisms found in natural water. Explain their effects on water quality.**

Answer:

Water available in nature may contain various types of living organisms. Some of them are described below:

i. Algae:

- These are the simple photosynthetic unicellular aquatic plants.
- They grow abundantly in nature, especially in the places of excreta disposal.
- The growth of algae can be controlled by adding copper sulphate or chlorine in the water.
- **Examples:** Green algae, diatoms and blue-green algae.

Effects:

- Produce taste and odour problems.
- Excessive growth causes algal blooms.
- Algal decomposition consumes DO and may cause fish mortality.
- Some cyanobacteria produce toxins harmful to humans and animals.

ii. Bacteria:

- Bacteria are unicellular organisms usually colorless and are the lowest form of life capable of synthesizing protoplasm from the surrounding environment.
- They are prokaryotic, unicellular, either free living in soil or water or parasites of plants or animals.
- The bacteria usually found in water range from 1 micron to 4 microns in length.
- **Examples:** E. coli, coliform bacteria, nitrifying bacteria, iron bacteria.

Effects:

- Decompose organic matter and reduce dissolved oxygen (DO).
- Some pathogenic bacteria cause diseases such as cholera, typhoid and dysentery.
- Coliform bacteria indicate possible fecal contamination.
- Some bacteria produce unpleasant taste and odor.

iii. Viruses:

- These are the smallest biological structures known to contain all the genetic information necessary for their reproduction.
- They can be defined as a group of infectious agents that are smaller than ordinary bacteria.
- These organisms have quite significant role in the spread of waterborne diseases.
- **Examples:** Hepatitis A virus, rotavirus, norovirus.

Effects:

- Cause waterborne diseases such as hepatitis, gastroenteritis and diarrhoea.
- Their presence indicates contamination, particularly from human sewage.
- They may survive in water and require effective disinfection for removal.

iv. Worms/Helminthes:

- The life cycles of worms or helminths often involve two or more animal hosts, one of which can be human and water contamination may result from human or animal waste that contains worms.
- These are microscopic as well as macroscopic and can enter directly to the human body through the skin or on the drinking of water.
- **Examples:** Ascaris, hookworm and tapeworm.

Effects:

- Some are transmitted through contaminated water and cause intestinal diseases.
- Their eggs and larvae may be resistant to ordinary disinfection.
- Their presence indicates contamination by human or animal excreta.

3. Write short notes on:

a. Infiltration gallery

b. Hydraulic grade line

c. Break pressure tank

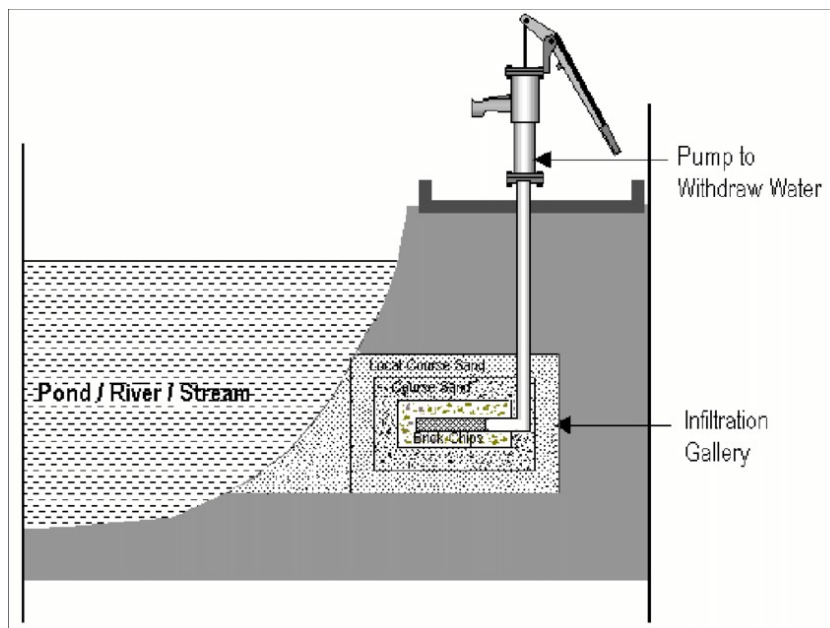
Answer:

a. Infiltration gallery:

An **infiltration gallery** is a **horizontal underground drainage structure** constructed below the ground surface, usually near or beneath a riverbed, to collect **groundwater that has infiltrated through permeable soil or river deposits**. It is commonly used as a source of water supply where shallow groundwater is available.

Construction:

- It generally consists of **perforated pipes or porous masonry/concrete galleries** laid horizontally.
- The gallery is surrounded by **graded filter materials** such as gravel and sand to prevent entry of fine soil particles.
- It is connected to a **collection well or sump**, from where water is pumped or conveyed to the treatment plant.
- It is usually placed **below the groundwater table**.



b. Hydraulic grade line:

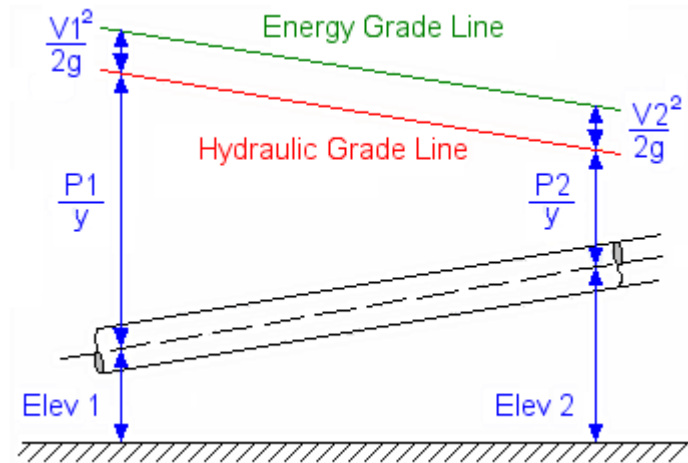
Hydraulic Grade Line (HGL) is an imaginary line joining the levels to which water would rise in piezometer tubes connected to a pipeline. It represents the sum of pressure head and elevation head of flowing water.

$$HGL = z + \frac{p}{\gamma}$$

where:

z = elevation head

$\frac{p}{\gamma}$ = pressure head



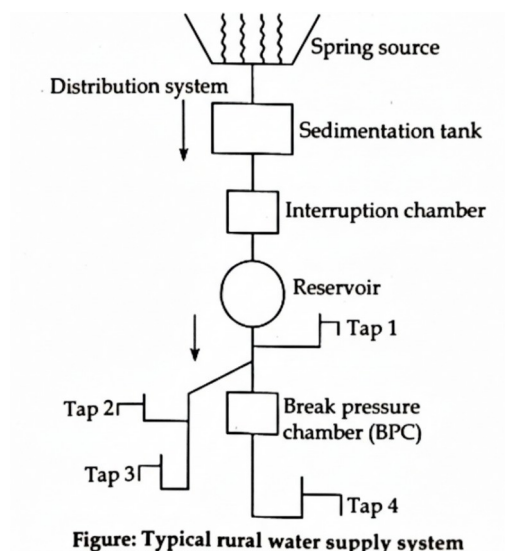
- HGL shows the variation of pressure along a pipeline.
- It always slopes downward in the direction of flow due to head loss from friction.
- In open channels, the HGL coincides with the water surface.
- In pressurized pipes, the HGL lies above the pipe centerline when the pressure is positive.

Use of HGL:

- Used in the design and analysis of water supply and pipe distribution systems.
- Helps determine the pressure available at different points in a pipeline.
- Identifies locations of low or negative pressure that may cause cavitation or air entry.
- Assists in selecting pump head and reservoir elevation.

c. Break pressure tank:

Break pressure tank is a pressure releasing device to break the pressure with float valve to prevent bursting of pipe due to excessive pressure, used in water supply systems, particularly in gravity flow schemes common in the hilly regions of Nepal. Its primary purpose is to "break" the static pressure in a pipeline, resetting it to atmospheric pressure to prevent pipes from bursting due to excessive force.



Key Characteristics and Mechanism

- **Pressure Regulation:** It reduces high hydrostatic pressure that builds up when there is a significant elevation difference between the water source and the distribution area.
- **Float Valve Operation:** Unlike an interruption chamber, which breaks pressure without a valve, a BPT typically utilizes a float valve to control the flow and release pressure effectively.
- **Placement:** These tanks are generally installed within the distribution mains of a water network.

Functions of Break Pressure Tank in pipe network:

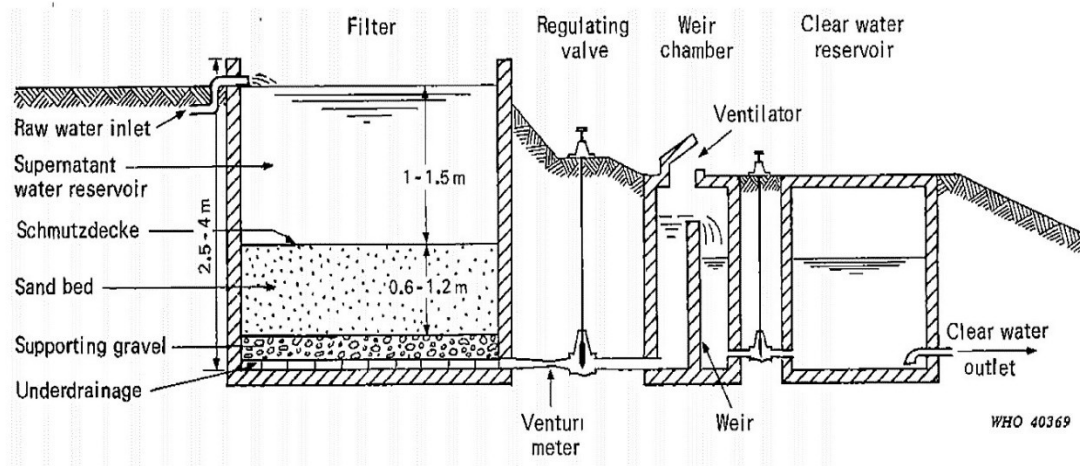
- **Prevents Structural Failure:** By keeping internal pressure within the rated capacity of the pipes, BPTs prevent frequent pipe bursts and joint failures.
- **Pressure Zoning:** In systems where elevation differences are large, BPTs are used to divide the network into different pressure zones. This ensures that while a minimum residual pressure (typically 7–10 meters for rural systems) is maintained at consumer taps, the pressure at lower elevations does not become dangerously high.
- **Head Loss Management:** While BPTs are used to manage pressure, engineers must still design the overall network to minimize friction and minor head losses to ensure efficient water delivery.

4. Differentiate between slow sand filter and rapid sand filter. Write down its merit and demerit.

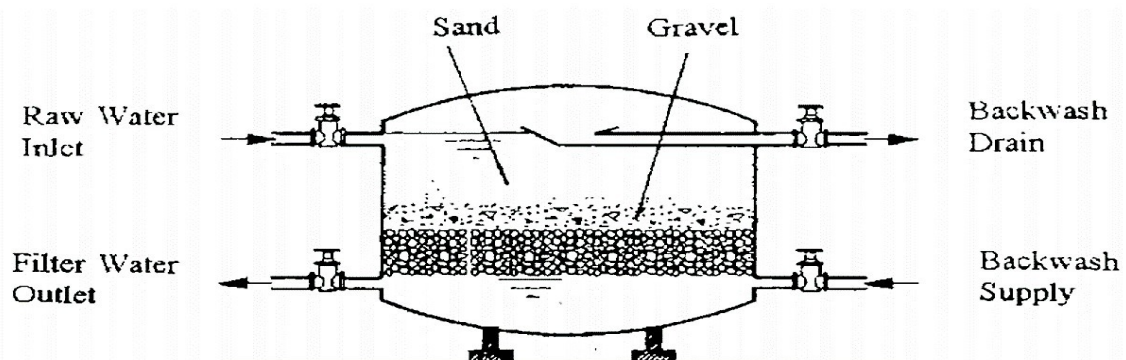
Answer:

The differences between slow sand filter and rapid sand filter are stated in points below:

Slow sand filter	Rapid sand filter
1. This type of filters has very slow rate of filtration hence called as slow sand filters.	1. This type of filters has high rate of filtration hence called as rapid sand filters.
2. Uneconomical and costly to install.	2. Economical and less costly to install.
3. Slow sand filter requires large area and large quantity of sand.	3. Rapid sand filter requires less area and small quantity of sand.
4. Water having low turbidity (less than 20 NTU) can be directly feed to slow sand filter without any pretreatment.	4. Pretreatment of raw water is required before passing the water through rapid sand filter.
5. Backwashing is not provided in slow sand filter.	5. Backwashing is provided in rapid sand filter which is done in every 1-3 days.
6. slow sand filter is highly efficient in removal of bacterial load.	6. Rapid sand filter is highly efficient in color removal bus less effective to remove bacterial load.



a. Slow sand Filtration



b. Rapid sand filtration

Merits of slow sand filter:

- SSF is highly efficient in removing bacterial loads, achieving higher purification through straining, adsorption, and biological action.
- It utilizes a biological layer known as the schmutzdecke, which develops on the sand surface and plays a major role in purifying the water.
- Water with low turbidity (less than 20 NTU) can be directly fed into a slow sand filter without the need for prior chemical treatment.
- The system has lower operational complexity, making it suitable for municipal schemes where space is available but technical oversight is limited.

Demerits of slow sand filters:

- These filters require a massive surface area and a large quantity of sand to function, making them difficult to implement in congested urban areas.
- Because of the substantial land and material requirements, they are often considered uneconomical and costly to install.
- As the name suggests, the rate of filtration is very slow, which limits the volume of water that can be treated per day.
- SSF is less effective at removing color from raw water compared to its rapid counterpart.
- Maintenance involves the physical scraping of the top layer of sand rather than an automated process.

Merits of rapid sand filter:

- RSF operates at a much higher speed, allowing for the treatment of large volumes of water in a short time.
- It is economical to install because it requires significantly less land and smaller quantities of sand than SSF.
- RSF is highly efficient at removing color and fine turbidity from water.
- These filters are cleaned through backwashing (using water and compressed air), a process that is faster and more efficient than manual scraping.

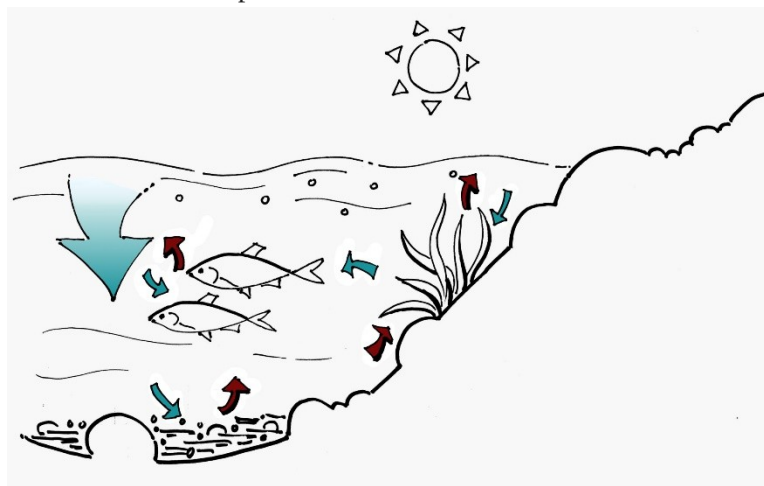
Demerits of rapid sand filter:

- While effective for solids, RSF is less effective at removing bacterial loads compared to SSF.
- Raw water must undergo pre-treatment, such as coagulation-flocculation and sedimentation, before it can be processed by a rapid sand filter.
- The high filtration rate can lead to technical issues such as air binding (trapped air blocking pores), the formation of mud balls, and cracking or damaging of the filter bed.
- The filter media must be cleaned very frequently, typically every 1 to 3 days, or whenever the head loss reaches 2.5 to 3 meters.

5. **What do you understand by self-purification of streams? Explain the various factors affecting self-purification.**

Answer:

When the sewage is discharged into a natural stream or river, the organic matter present in the sewage gets oxidized by bacteria and converted into simple, inoffensive, stable substances. In this process of oxidation, the dissolved oxygen content of river or stream is utilized and the deficiency of dissolved oxygen is created. This deficiency thus created is filled up by the absorption of atmospheric oxygen. Thus, dissolved oxygen of stream water is consumed by sewage for the oxidation purpose and at the same time oxygen is consumed from atmosphere, this process is called self-purification of river/stream.



The various factors affecting self-purification are as follows:

i. Dilution:

- When sewage is discharged into large volume of water flowing in a natural stream or river, it is dispersed and dilution takes place.
- Due to dilution, the concentration of organic matter, BOD, suspended solids are reduced.
- The concentration of resulting mixture of sewage and water is given by:

$$C = \frac{Q_s C_s + Q_w C_w}{(Q_s + Q_w)}$$

Where:

C = concentration of resulting mixture
Q_s = flow of sewage
C_s = concentration of pollutant in sewage
Q_w = flow of water
C_w = concentration of pollutant in water

ii. Oxidation:

- When sewage is discharged in water body, the organic matter present in the sewage is oxidized by aerobic bacteria utilizing dissolved oxygen of river/stream.
- The deficiency of oxygen is filled up from atmospheric oxygen.
- Oxidation continues till the oxidation demand is fully satisfied and the river/stream is completely purified.

iii. Reduction:

- Anaerobic bacteria split complex organic matter into liquids and gases and gets settled at the bottom.
- This reduction of organic matter helps for the process of oxidation and hence assists the process of self-purification.

iv. Sedimentation:

- The settleable solids contained in sewage settle at the bottom of streams and easily separated.
- Anaerobic decomposition takes place in settleable solids.
- Hence, Sedimentation process helps in self-purification of water.

v. Sunlight:

- It has bleaching and stabilizing effect on bacteria.
- It acts as disinfectant and stimulates the growth of algae which produce oxygen in daylight and utilizes at night.
- Hence, all such effects of sunlight assist in self-purification of water.

vi. Current:

- Heavy current helps in thorough mixing of sewage.
- Slow current promotes algae growth and sedimentation.
- Both heavy and slow current are necessary for self-purification purpose.

6. Classify the sedimentation tanks. How can we increase the settling efficiency of particles?

Answer:

A **sedimentation tank** is a unit in water and wastewater treatment in which suspended particles are removed by allowing them to settle under gravity.

i. Based on the method of operation:

a. Intermittent or fill-and-draw tank

- Water is admitted and allowed to settle for a specified period.
- After settling, clarified water is withdrawn.

- Sludge is then removed.
- Suitable for **small water-treatment plants**.

b. **Continuous-flow tank**

- Water continuously enters the tank.
- Settled sludge is continuously or periodically removed.
- Clarified water continuously leaves the tank.
- Commonly used in **large treatment plants**.

ii. Based on shape:

a. **Rectangular sedimentation tank**

- Long rectangular basin with horizontal flow.
- Easy to construct and operate.
- Common in water-treatment plants.

b. **Circular sedimentation tank**

- Long rectangular basin with horizontal flow.
- Easy to construct and operate.
- Common in water-treatment plants.

iii. Based on purpose:

- **Plain sedimentation tank:** Removes settleable suspended solids without chemical addition.
- **Sedimentation after coagulation:** Removes **flocs** formed during chemical treatment.
- **Secondary sedimentation tank:** Used after biological wastewater treatment to separate biological solids from treated effluent.

The settling efficiency of particles can be increased by:

- Increasing detention time so that particles have sufficient time to settle.
- Reducing the surface overflow rate, allowing more particles to settle.
- Providing proper inlet and outlet arrangements to ensure uniform flow and prevent short-circuiting.
- Using coagulation and flocculation to convert fine particles into larger, heavier flocs.
- Increasing the effective settling area, such as by using tube settlers or plate settlers.
- Preventing turbulence inside the tank because turbulence can resuspend settled particles.

7. **What are preliminary and secondary treatment process? Describe the working principle of activated sludge process.**

Answer:

Sewage treatment is a sequential process designed to remove impurities and transform wastewater into a stable effluent that is safe for environmental disposal. The treatment is broadly categorized into preliminary, primary, and secondary stages to progressively target different types of contaminants.

Preliminary Treatment Process:

The preliminary stage focuses on physical unit operations that use physical forces to remove large floating objects and heavy inorganic solids that could otherwise damage or clog downstream equipment.

- **Intake Structures:** These hydraulic devices draw water from the source and are equipped with gates to prevent the entry of large debris and heavy bed loads into the treatment system.
- **Screening:** Sewage is passed through coarse, medium, or fine screens to trap and remove floating matter such as sticks, branches, plastic, and even dead animals. This essential step reduces the organic load on subsequent units and improves overall treatment efficiency.
- **Grit Chamber:** This unit utilizes differential sedimentation to separate heavy inorganic materials like sand, gravel, and grit from the lighter organic matter, which remains in suspension.

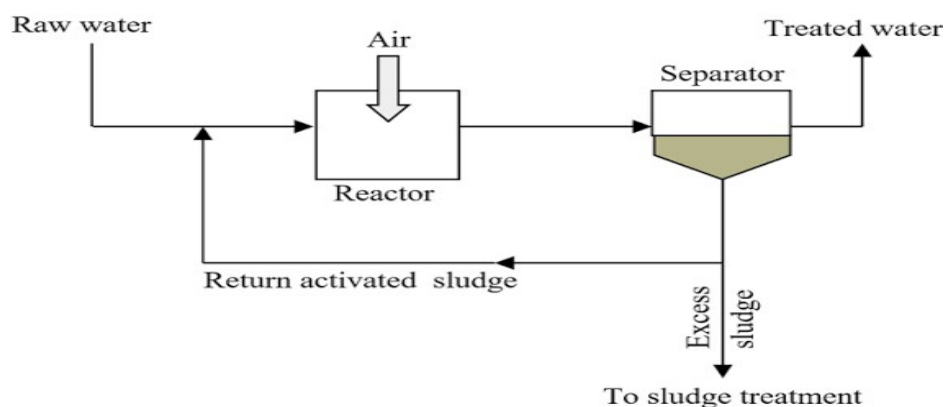
- **Skimming:** This process involves removing oils, grease, and fats from the sewage. Air is typically blown from the bottom of a rectangular tank, causing bubbles to coagulate with the grease, making it rise to the surface for manual or mechanical removal.

Secondary Treatment Process

Secondary treatment, also known as biological treatment, primarily utilizes microorganisms to decompose and remove dissolved and colloidal organic matter that remains after preliminary and primary stages.

- **Activated Sludge Process:** This is an aerobic biological method that maintains a mixture of bacteria, fungi, and protozoa in suspension within an aeration tank.
 - **Mechanism:** An aerator transfers oxygen into a "mixed liquor" (influent and activated sludge) to oxidize organic matter, converting it into biological flocs.
 - **Efficiency:** ASP is highly effective, typically achieving BOD removal efficiencies of 95% to 98%.
- **Trickling Filters:** This is an attached-growth aerobic process where wastewater is distributed over a bed of filter media covered in a microbial biofilm.
 - **Mechanism:** As the sewage trickles through the media, organic matter is oxidized by the microorganisms on the film.
 - **Efficiency:** It generally removes between 75% to 90% of BOD and suspended solids.
- **Oxidation Ponds:** These are large, shallow earthen basins where stabilization occurs through the combined action of algae and various types of bacteria (aerobic, anaerobic, and facultative). They are characterized by long detention periods ranging from several days to weeks.
- **Secondary Settling Tank:** This final stage of secondary treatment involves liquid-solid separation. The biological cell tissue formed in previous steps settles to the bottom by gravity, leaving a clarified effluent safe for disinfection and final discharge.

Activated sludge process is a biological method of wastewater treatment that utilizes a mixture of microorganisms such as bacteria, fungi, protozoa and rotifers maintained in suspension, in an aerobic environment to metabolize organic matter to produce secondary wastewater effluent of acceptable quality. It has removal efficiencies in the range of 95-98% than trickling filters of 80-85%.



The main stages involved in activated sludge process is described below:

i. Physiochemical adsorption:

The fine suspended particles of the sewage combine with bacteria to form a sub layer of bacterial film at the surface. The film attracts finely divided colloidal and dissolved organic matter and result in formation of flocs. The flocs get settled at the bottom of sedimentation tanks and are removed.

ii. Oxidation:

Large flocs formed act as a vehicle for aerobic bacteria that oxidizes the carbonaceous and organic matter of wastewater by biochemical action. Settled microorganisms are removed from the bottom of

settling tank are and are returned to aeration tank to metabolize additional organic matter. Balance between food and mass of the organisms, also known as sludge loading ratio must be maintained.

Working:

- A primary clarifier may be introduced to remove part of suspended solid present in the influent and this reduces the organic load to the activated sludge system.
- The biological aerator is filled with a mixer of activated sludge and influent known as mixed liquor. It is necessary to maintain certain liquid suspended solid (MLSS) in the aerated tank which maintain good removal efficiency.
- The aerator equipment transfers oxygen into the reactor for the oxidation of organic matter.
- The continuous introduction of new influent results in a continuous discharge of mixed liquor to the secondary settling tank where separation of solid and liquid takes place.
- The liquid leaves the system as a treated effluent whereas some part of this sludge is recirculated to aeration tank called as return sludge and rest is taken for anaerobic digestion.

8. Why are community mobilization and participation important? Define special role of women participation in WSP.

Answer:

Community participation in the water and sanitation sector is a process where local beneficiaries, community organizations, and stakeholders actively engage in every stage of a project, including planning, design, implementation, operation, maintenance, and monitoring. This participatory approach ensures that the infrastructure developed meets the actual needs and priorities of the community, fostering long-term utilization and sustainability.

Community mobilization and participation are important because of several reasons:

- i. **Sense of Ownership:** Early and active involvement creates a deep sense of ownership among users, making them more likely to protect and contribute to the system's longevity.
- ii. **Conflict Resolution:** Committees facilitate community consensus on project sites and canal/pipeline alignments, which minimizes future disputes regarding land acquisition, right-of-way, and water source rights.
- iii. **Resource Mobilization:** User committees effectively mobilize local resources, such as voluntary labor, cash contributions, and locally available raw materials, which significantly reduces construction and maintenance costs.
- iv. **Efficient Operation and Maintenance (O&M):** Local management allows for quick repairs of leaks or faults that might otherwise be delayed if managed by a distant government agency.
- v. **Indigenous Knowledge:** Integrating local farmers' knowledge of the terrain, soil, and river behavior leads to more pragmatic and cost-effective technological solutions.
- vi. **Sustainability and Transparency:** Committees are responsible for collecting tariffs and managing bank accounts, ensuring revenue is available for O&M while promoting financial transparency and accountability.

WSP (Water Safety Plan) is a systematic approach to ensure the safety of drinking water from its source to the consumer. Women have a special and important role in WSP as they are often the primary users, managers, and collectors of household water.

Roles of women in WSP include:

- i. **Water quality monitoring:** Women can identify changes in water color, taste, odor, and cleanliness and report contamination.
- ii. **Hygiene promotion:** Women can promote handwashing, safe food preparation, proper sanitation, and hygienic water handling within households.

- iii. **Community participation:** Women can participate in WSP committees, planning, decision-making, monitoring, and implementation.
- iv. **Awareness and education:** They can educate family members and communities about safe drinking-water practices.
- v. **Sustainability:** Women's involvement improves community ownership, accountability, and long-term sustainability of water-supply systems.

9. **a. What are the different onsite sanitation systems practiced in Nepal?**

b. Describe the principles and benefits of ecological sanitation.

Answer:

a. On-site sanitation is a system in which human excreta and wastewater are collected, treated, stored and/or disposed of at or near the place where they are generated, rather than being conveyed through a centralized sewer network.

The major on-site sanitation systems practiced/promoted in Nepal are:

- i. **Pit latrine:**
 - Excreta is collected in a pit constructed below or near the latrine.
 - It may be a single-pit or double-pit system.
 - Common in rural and low-income areas.
- ii. **Ventilated Improved Pit (VIP) latrine:**
 - An improved pit latrine provided with a ventilation pipe.
 - The pipe helps reduce odor and flies.
 - Suitable where water availability is limited.
- iii. **Pour-flush latrine:**
 - Excreta is flushed manually using a small quantity of water.
 - It may discharge into a pit, septic tank or other containment system.
 - Widely used in Nepal, particularly where piped water supply is limited.
- iv. **Septic tank system:**
 - Toilet wastewater is conveyed to a septic tank, where solids settle and anaerobic decomposition occurs.
 - The liquid effluent is generally disposed of through a soak pit/leach field.
 - It is particularly common in urban and peri-urban residential areas.
- v. **EcoSan (Ecological Sanitation) / urine-diversion toilet:**
 - Human excreta are collected separately and treated/recovered for agricultural reuse.
 - Urine may be used as fertilizer after appropriate treatment/dilution.
 - EcoSan has been promoted in different parts of Nepal, although its implementation remains relatively limited.
- vi. **Composting toilet:**
 - Faecal matter is biologically decomposed and converted into a compost-like material.
 - It requires little or no flushing water and is useful where water is scarce.
 - Nepal's sanitation classifications include composting toilets among on-site facilities.

b. **Ecological Sanitation also known as EcoSan**, is a sustainable sanitation approach that **safely treats and reuses human excreta as a resource** rather than treating it as waste. It aims to protect public health, conserve water, and recycle nutrients back to the soil. Ecological sanitation approach is suitable for rural communities, water-scarce areas, schools and institutions. According to UNICEF, 40 out of 100 people in Nepal have access to safely managed on-site sanitation and about 53% use safely managed sanitation services with roughly 62% to provide general improved facilities nationwide.

The examples of EcoSan are Urine Diverting Dry toilets (UDDT), Composting toilets, Double-vault Ecosan toilets. In Nepal, Ecological sanitation approach is actively promoted by organizations like the Environment and Public Health Organisation (ENPHO) and WaterAid in Nepal. EcoSan double-vault and urine-diverting toilets are mostly concentrated in agricultural peri-urban and rural clusters such as Siddhipur, Lubhu, and Kirtipur in the Kathmandu Valley.

Advantages:

- Conserves water by reducing or eliminating flushing.
- Recycles nutrients such as nitrogen, phosphorus, potassium for agriculture.
- Reduces environmental pollution.
- Improves soil fertility and crop production.
- Suitable for water-scarce and rural areas.
- Low operation and maintenance costs.

Principles of ecological sanitation:

- i. **Prevention of pollution:**
 - Prevents contamination of surface water, groundwater and soil by human excreta.
 - Reduces discharge of untreated wastewater into the environment.
- ii. **Separation of urine and feces:**
 - Urine and feces may be collected separately because they have different characteristics and treatment requirements.
 - Urine is relatively rich in nutrients, especially nitrogen and phosphorus.
- iii. **Safe treatment of excreta:**
 - Human excreta must be properly treated to destroy pathogens, parasites and harmful organisms before reuse.
 - Methods include dehydration, composting and controlled storage.
- iv. **Resource recovery and reuse:**
 - Nutrients from human excreta are recovered and reused as fertilizer or soil conditioner.
 - Treated wastewater can potentially be reused for irrigation or other suitable purposes.
- v. **Water conservation:**
 - EcoSan systems generally aim to minimize or eliminate the use of water for transporting feces.
 - This reduces freshwater consumption and wastewater generation.

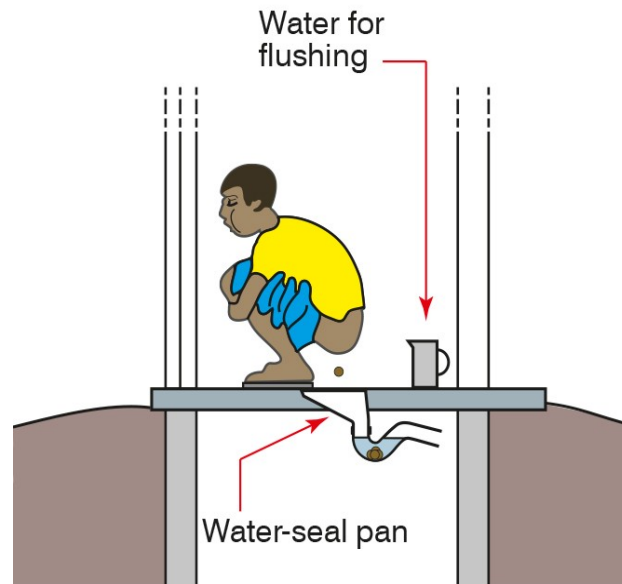
10. Describe pour-flush toilet and VIP toilet with neat sketches.

Answer:

Pour flush toilet is used for the disposal of excreta by providing water seal which does not let smell to escape into surroundings.

Construction of pour-flush toilets:

- i. A circular shaped pit of 1.5 m depth is dug for stability.
- ii. The pit is built of brick or stone masonry covered with RCC slab.
- iii. A squatting plate or slab made of concrete with water seal pan is kept under superstructure. From water seal pan, a pipe with sufficient grade is connected to the junction chamber.
- iv. The junction chamber, built of brick or stone masonry is covered by RCC slab and connected to waste pipe.
- v. Waste pipe is fixed at 15° slope to horizontal. The diameter of PVC is 10 cm. It should be extended at least 10 cm to prevent liquid damaging the wall of pit.
- vi. The junction box has two functions: to divert waste flow to either pit, to allow access to clear blockage in waste pipe.



Advantages:

- i. Reduction of flies, mosquitoes and odor.
- ii. Easy to clean.
- iii. Easy to construct and maintain.

Disadvantages:

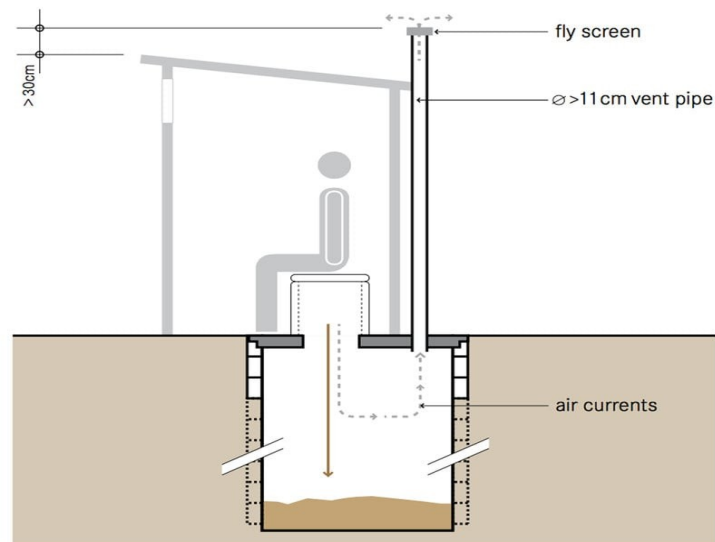
- i. It requires regular water for operation.
- ii. Water seal prevents the use of solid anal cleaning materials.
- iii. Expensive than other types of latrines.
- iv. High skill to construct the pan.

VIP toilets:

- **VIP toilet** refers to a **Ventilated Improved Pit (VIP) latrine**, a type of improved pit toilet designed to control **odor, flies, and other nuisances**.
- The main purpose of VIP toilets is to overcome the unhygienic conditions of pit privy latrine.

Construction steps:

- i. It has four major compartments: pit slab, superstructure and vent pipe.
- ii. It consists of pit with minimum 1.3 m by 1 m in plan and 1.5 to 2.8 m in depth.
- iii. It is laid at least 30 m from ground water.
- iv. Laying, casting and curing of slabs are done.
- v. Vent pipes and cover slabs are tested for strength.
- vi. The pit is excavated and ring beam is constructed along with installation of pit covers.
- vii. The superstructure of length 120 cm, width 80 cm, height 100 cm and 180 cm are constructed.
- viii. Finally vent pipes and seat slabs are installed.



Advantages:

- i. Control over odors and insects.
- ii. Easily built and repaired with locally available materials.
- iii. Does not require constant water source.
- iv. It offers long term solution.

Disadvantages:

- i. Risk of groundwater pollution.
- ii. Manual removal of human faeces is required.
- iii. Lack of space for relocating pit in densely populated areas.